

Spikeling Exploration Activity

Objective

Before we begin the guided tutorial, spend some time exploring how Spikeling behaves. Your goal is to develop intuition about how neurons respond to different kinds of stimulation.

For each experiment:

- Change **only one parameter at a time** whenever possible.
 - Observe both the **LED activity** and any plots or sounds produced by Spikeling.
 - Record what you notice before moving on.
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1. Stimulus Strength

Keep everything else constant.

- Start with a very weak stimulus.
- Gradually increase the stimulus strength.
- Continue increasing until the neuron is firing rapidly.

Observe

- When does the neuron begin to spike?
 - Does the firing rate change as stimulus strength increases?
 - Is there a point where increasing the stimulus has little additional effect?
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2. Stimulus Frequency

Keep the stimulus amplitude constant.

- Deliver slow pulses.
- Gradually increase the frequency of stimulation.

Observe

- Can the neuron follow every pulse?
- Does firing become more continuous at higher frequencies?
- Are there frequencies where the neuron cannot keep up?

3. Membrane Voltage (Excitability)

Adjust the membrane voltage (or resting potential) while keeping the stimulus the same.

Try making the neuron:

- less excitable
- more excitable

Observe

- Does it become easier or harder to generate spikes?
 - How does the firing threshold change?
 - Does spontaneous activity appear?
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4. Noise

Increase and decrease the noise level.

Observe

- Does the neuron ever spike without an obvious stimulus?
 - Does noise make the firing pattern more or less predictable?
 - Can weak stimuli trigger spikes when noise is present?
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5. Different Neuron Types

Switch between the different neuron models available in Spikeling.

For each neuron:

- Apply the same stimulus.
- Compare the responses.

Observe

- Which neuron fires most easily?

- Which fires in bursts?
 - Which adapts during sustained stimulation?
 - Which responds only at the beginning of a stimulus?
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6. Photoreceptor

The photoreceptor converts light into an input that can drive the neuron.

The **Photoreceptor Gain** controls how strongly changes in light affect the neuron.

Experiment A: Light Intensity

Keep the photoreceptor gain constant.

- Shine the LED on the photoreceptor.
- Change the Stimulus Strength

Observe

- How does changing the amount of light affect the neuron's activity?
 - Does brighter light always produce more spiking?
 - Is the response immediate or gradual?
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Experiment B: Photoreceptor Gain

Keep the LED at a fixed strength.

- Start with low photoreceptor gain.
- Gradually increase the gain.
- Repeat the same light changes at different gain settings.

Observe

- How does increasing the gain change the neuron's response?
 - Does the neuron become more sensitive to small changes in light?
 - At very high gain, does the neuron become overly responsive?
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8. Build Your Own Experiment

Design one experiment that interests you.

Examples:

- What happens if you combine strong stimulation with high noise?
- Can you make the neuron fire rhythmically?
- Can you stop it from firing altogether?
- Can you predict what will happen before you test it?

Record:

- Your question
 - What you changed
 - What you observed
 - Whether the results matched your prediction
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Reflection Questions

After exploring, discuss with your group:

1. Which parameter had the biggest effect on firing?
 2. What determines whether the neuron spikes or not?
 3. Which neuron type behaved most differently from the others?
 4. What surprised you the most?
 5. If this were a real neuron, what biological property do you think each control represents?
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General Tips

- Change only one variable at a time.
- Make predictions before each experiment.
- If something unexpected happens, try to explain it rather than assuming it's an error.
- There are no "wrong" observations—your goal is to build intuition about neuronal dynamics.

This format encourages inquiry-based learning while naturally introducing concepts such as threshold, firing rate, adaptation, excitability, bursting, sensory transduction, and noise, which the formal tutorial can then explain in more detail.